LESSON 4.1

Understand Percent

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can explain a percent as a rate per 100, model it on a decimal grid or tape diagram, and interpret percents greater than 100%.

WHY THIS MATTERS The arcade posts every result as a percent of the weekly goal. A percent compares a number to 100 — so 40% means 40 out of every 100. Some players land under the goal, and some blow past it: a score of 150% means one and a half times the whole goal. Your job is to read and model each result as a comparison to 100.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Percent *Spanish: Porcentaje*	A ratio that compares a number to 100. The word means “per hundred,” and the symbol is %.	40% = 40 out of every 100
Per hundred *Spanish: Por cada cien*	The meaning hidden inside the word percent: for every group of 100. It is why every percent can be written as a fraction over 100.	7% means 7 per hundred
Decimal grid *Spanish: Cuadrícula decimal*	A 10-by-10 grid of 100 squares. Shading squares shows a percent directly, one square per percent.	62 shaded squares = 62%
Part-to-whole *Spanish: Parte a todo*	A comparison of one part against the whole amount. In a percent, the whole is always 100%.	shaded squares to all squares
Greater than 100% *Spanish: Mayor que 100 %*	More than one whole. A percent above 100 means the amount is larger than the thing it is being compared to.	150% of the gray kitten = 1.5 times its weight
Equivalent ratio *Spanish: Razón equivalente*	A ratio with the same value, made by multiplying or dividing both terms by the same number.	2 : 5 → 40 : 100

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

3 out of 4 students bring lunch. What percent?

1. What times 4 makes 100? 25.
2. 3 × 25 = 75 and 4 × 25 = 100.
3. 75 out of 100 = 75%.

Answer: 75% bring lunch

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Shanice's gray kitten weighs 4 ounces. The orange kitten weighs 1.5 times as much. What percent of the gray kitten's weight is the orange kitten's weight?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- _____ % means _____ out of every hundred.
- To compare to a hundred, I multiplied both terms by _____ .
- The part is _____ and the whole is _____ .
- A percent is a ratio because _____ .

Word bank percent | per hundred | ratio | part | whole | equivalent | multiply | terms | grid | compare

WATCH OUT!

A common mistake

A common mistake with percent is scaling only one term of the ratio: reading 7 out of 10 as 7%. A percent compares to 100, so BOTH terms must be multiplied by the same factor

LESSON 4.1

Understand Percent

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

What does the word percent mean?

(A) Per ten
(B) Per thousand
(C) Per hundred
(D) Per part

HINT Look at the second half of the word: cent.

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Sort each percent by how it compares to one whole.

Groups: Less than one whole | More than one whole

_____ 45%

_____ 150%

_____ 99%

_____ 200%

_____ 0.5%

_____ 101%

HINT What percent equals one whole?

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

3 out of every 10 arcade players reach the bonus level. What percent is that?

(A) 3%
(B) 10%
(C) 13%
(D) 30%

HINT A percent compares to 100, so rewrite the ratio out of 100.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A decimal grid has 100 squares and 65 are shaded. What percent is shaded?

(A) 6.5%
(B) 35%
(C) 65%
(D) 100%

HINT How many squares are in the whole grid?

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Which percent means MORE than one whole?

(A) 0.8%
(B) 80%
(C) 100%
(D) 120%

HINT What percent stands for one whole?

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Find the error

1. **Setup** 7 out of every 10 seats are taken. What percent is that?
2. **Step 1** A percent compares to 100.
3. **Step 2** 7 out of 10 = 7%.
4. **Step 3** So 7% of the seats are taken.

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Does 7 out of 10 mean mostly full or mostly empty?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

A friend says "a percent can never be more than 100, because you cannot have more than all of something." How would you answer using the mascot vote and the kittens?

SENTENCE STARTER A percent compares a number to ___, so ___% just means the amount is ___ than the whole it is compared to.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 4.1

Understand Percent

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Write each ratio as an equivalent ratio out of 100, then as a percent.

Ratio	Out of 100	Percent
4 out of 10	40	40%
7 out of 20		
3 out of 4		

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each comparison to its percent.

_____ 4 out of 10

_____ 23 out of 100

_____ 1.5 times the whole

_____ the whole itself

(A) 40%

(B) 23%

(C) 150%

(D) 100%

3 **Mark and label each value on the number line.**

Place these percents on the line: 25%, 100%, and 150%. Mark where one whole sits.



SHOW YOUR THINKING

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

The orange kitten weighs 1.5 times as much as the gray kitten. What percent of the gray kitten's weight is that?

(A) 1.5%

(B) 15%

(C) 50%

(D) 150%



0% 50% 100%

WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Find the error

1. **Setup** A player scored 150% of the weekly goal of 200 points.
2. **Step 1** 150% is more than one whole.
3. **Step 2** So the player scored 150 points.
4. **Step 3** The player beat the 200-point goal.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

A friend says "a percent can never be more than 100, because you cannot have more than all of something." How would you answer using the mascot vote and the kittens?

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LESSON 4.1

Understand Percent

Challenge · Extension

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- ☐ Exceeds
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the table comparing each amount to one whole.

Percent	Compared to one whole	Times the whole
50%	less	0.5
100%		
150%		
200%		

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the error

1. **Setup** Which is larger: 40% of 200, or 150% of 40?
2. **Step 1** 150% is greater than 40%.
3. **Step 2** So 150% of 40 must be the larger amount.
4. **Step 3** 150% of 40 is larger.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A car dealership sold 50 cars last month: 20 sedans, 15 SUVs, 10 trucks, and 5 vans. The owner will order 100 cars for next month and wants the order to match last month's sales pattern. Write each type as a percent, then say how many of each to order — and explain why percents make this easier than the raw counts.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

A friend says "a percent can never be more than 100, because you cannot have more than all of something." How would you answer using the mascot vote and the kittens?

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